# Berkovich Space

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# 1 Spectrum

Let $\mathbf{k}$ be a spherically complete non-archimedean field which is algebraically closed and $A = \mathbf{k}[T]$. We want to consider the "analytic structure" on $\operatorname{mSpec} A$. However, unlike the complex case, the set $\operatorname{mSpec} A$ is totally disconnected with respect to the topology induced by the absolute value on $\mathbf{k}$ (**??**). To overcome this difficulty, Berkovich uses multiplicative semi-norms to "fill in the gaps" between the points in $\operatorname{mSpec} A$, leading to the notion of the spectrum of a Banach ring.

## 1.1 Definition and examples

**Definition 1.1.** Let $R$ be a Banach ring. The *Berkovich spectrum* $\mathscr{M}(R)$ of $R$ is defined as the set of all multiplicative semi-norms on $R$ that are bounded with respect to the given norm on $R$. For every point $x \in \mathscr{M}(R)$, we denote the corresponding multiplicative semi-norm by $|\cdot|_x$.
We equip $\mathscr{M}(R)$ with the weakest topology such that for each $f \in R$, the evaluation map $\mathscr{M}(R) \to \mathbb{R}_{\geq 0}$, defined by $x \mapsto |f|_x =: f(x)$, is continuous.

**Example 1.2.** Let $(\mathbf{k}, |\cdot|)$ be a complete valuation field. The Berkovich spectrum $\mathscr{M}(\mathbf{k})$ consists of a single point corresponding to the given absolute value $|\cdot|$ on $\mathbf{k}$.

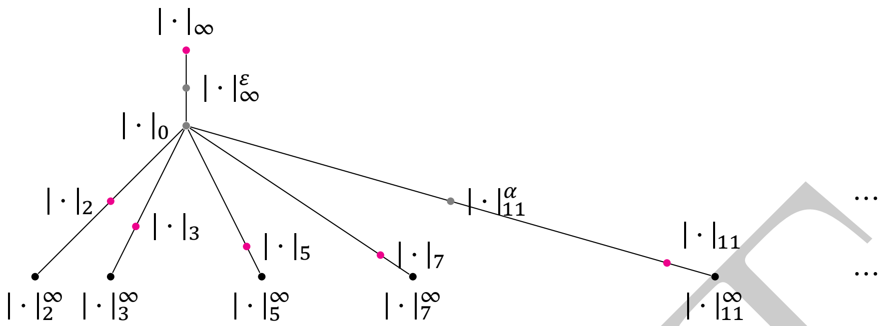
**Example 1.3.** Consider the Banach ring $(\mathbb{Z}, \|\cdot\|)$ with $\|\cdot\| = |\cdot|_\infty$ is the usual absolute value norm on $\mathbb{Z}$. Let $|\cdot|_p$ denote the $p$-adic norm for each prime number $p$, i.e., $|n|_p = p^{-v_p(n)}$ for each $n \in \mathbb{Z}$,

**Date**: February 24, 2026, **Author**: Tianle Yang, My Homepage

where $v_p(n)$ is the $p$-adic valuation of $n$. The Berkovich spectrum

$$\mathscr{M}(\mathbb{Z}) = \{|\cdot|_\infty^\varepsilon : \varepsilon \in (0,1]\} \cup \{|\cdot|_p^\alpha : p \text{ is prime}, \alpha \in (0,\infty]\} \cup \{|\cdot|_0\},$$

where $|a|_p^\infty := \lim_{\alpha\to\infty} |a|_p^\alpha$ for each $a \in \mathbb{Z}$ and $|\cdot|_0$ is the trivial norm on $\mathbb{Z}$.




Yang: To be continued.

**Theorem 1.4.** Let $R$ be a Banach ring. The spectrum $\mathscr{M}(R)$ is nonempty.

*Proof.* Let $\mathfrak{m}$ be a maximal ideal of $R$. Note that the pullback of residue norm on the residue field $R/\mathfrak{m}$ is bounded with respect to the given norm on $R$. Replacing $R$ by the completion of $R/\mathfrak{m}$, we may assume that $R$ is a complete field. Consider the set

$$\Sigma = \{\text{norm on } R \text{ bounded by the given norm } \|\cdot\|\}$$

with the partial order defined by boundedness. Since for a descending chain in $\Sigma$, the infimum is a norm, by Zorn's lemma, there exists a minimal element $|\cdot| \in \Sigma$.

We claim that $|\cdot|$ is multiplicative. Since the spectral radius $\rho(f) = \lim_{n\to\infty} |f^n|^{1/n}$ associated to $|\cdot|$ is power-multiplicative and bounded by $|\cdot|$, by minimality of $|\cdot|$, we have $\rho(f) = |f|$ for each $f \in R$. Thus $|\cdot|$ is power-multiplicative. If $|\cdot|$ is not multiplicative, then there exist $a, b \in R \setminus \{0\}$ such that $|ab| < |a||b|$. Then $|b| \leq |a^{-1}||ab| < |a^{-1}||a||b|$, which implies that $|a||a^{-1}| > 1$. Set $r = |a|^{-1} < |a^{-1}|$ and consider $R\langle T/r\rangle$. Since $r \cdot |a| = 1$, we have that

$$\left|\sum_{n=0}^\infty a^n T^n\right| = \sum_{n=0}^\infty |a^n| r^n = \sum_{n=0}^\infty |a|^n r^n = \sum_{n=0}^\infty 1 = \infty.$$

The power series is not convergent in $R\langle T/r\rangle$ and hence $1 - aT$ is not invertible in $R\langle T/r\rangle$. Let $\mathfrak{n}$ be a maximal ideal of $R\langle T/r\rangle$ containing $1 - aT$. Consider $R \to R\langle T/r\rangle \to R\langle T/r\rangle/\mathfrak{n}$. Since $R$ is a field, the composition is injective. The residue norm on $R\langle T/r\rangle/\mathfrak{n}$ induces a norm $|\cdot|'$ on $R$ bounded by $|\cdot|$. Note that $|a^{-1}|' \leq |T| = r = |a|^{-1} < |a^{-1}|$, contradicting the minimality of $|\cdot|$. $\square$

**Definition 1.5.** Let $\varphi : R \to S$ be a bounded ring homomorphism of Banach rings. The *pullback* map $\mathscr{M}(\varphi) : \mathscr{M}(S) \to \mathscr{M}(R)$ is defined by $\mathscr{M}(\varphi)(x) = x \circ \varphi : f \mapsto |\varphi(f)|_x$ for each $x \in \mathscr{M}(S)$.

Note that $\mathscr{M}(\varphi)(f^{-1}(V)) = \varphi(f)^{-1}(V)$ for each $f \in R$ and open subset $V \subset \mathbb{R}_{\geq 0}$. Hence the pullback map is continuous.

**Notation 1.6.** Let $R$ be a Banach ring and $x \in \mathscr{M}(R)$. We denote by $|\cdot|_x$ the multiplicative semi-norm on $R$ corresponding to the point $x$. Its kernel $\{f \in R : |f|_x = 0\}$ is a closed prime ideal of $R$, denoted by $\wp_x$.

**Definition 1.7.** Let $R$ be a Banach ring. For each $x \in \mathscr{M}(R)$, the *completed residue field* at the point $x$ is defined as the completion of the residue field $\kappa(x) = \mathrm{Frac}(R/\wp_x)$ with respect to the multiplicative norm induced by the semi-norm $|\cdot|_x$, denoted by $\mathscr{H}(x)$.

**Example 1.8.** Consider the Banach ring $(\mathbb{Z}, |\cdot|_\infty)$ as in Example 1.3. We have

- $x = |\cdot|_\infty^\varepsilon$ for some $\varepsilon \in (0,1]$: $\wp_x = (0)$ and $\mathscr{H}(x) \cong \mathbb{R}$ with the absolute value norm raised to the power $\varepsilon$;
- $x = |\cdot|_0$: $\wp_x = (0)$ and $\mathscr{H}(x) \cong \mathbb{Q}$ with the trivial norm;
- $x = |\cdot|_p^\alpha$ for some prime number $p$ and $\alpha \in (0,\infty)$: $\wp_x = (0)$ and $\mathscr{H}(x) \cong \mathbb{Q}_p$ with the $p$-adic norm raised to the power $\alpha$;
- $x = |\cdot|_p^\infty$ for some prime number $p$: $\wp_x = (p)$ and $\mathscr{H}(x) \cong \mathbb{F}_p$ with the trivial norm.

**Definition 1.9.** Let $R$ be a Banach ring. The *Gel'fand transform* of $R$ is the bounded ring homomorphism

$$\Gamma : R \to \prod_{x \in \mathscr{M}(R)} \mathscr{H}(x), \quad f \mapsto (f(x))_{x \in \mathscr{M}(R)},$$

where the norm on the product $\prod_{x \in \mathscr{M}(R)} \mathscr{H}(x)$ is given by the supremum norm.

**Example 1.10.** Let $K$ be a number field equipped with Yang: To be continued.

**Proposition 1.11.** The Gel'fand transform $\Gamma : R \to \prod_{x \in \mathscr{M}(R)} \mathscr{H}(x)$ of a Banach ring $R$ factors through the uniformization $R^u$ of $R$, and the induced map $R^u \to \prod_{x \in \mathscr{M}(R)} \mathscr{H}(x)$ is an isometric embedding. Yang: To be checked.

*Proof.* Yang: To be added. □

**Lemma 1.12.** Let $\{K_i\}_{i \in I}$ be a family of completed fields. Consider the Banach ring $R = \prod_{i \in I} K_i$ equipped with the product norm. The spectrum $\mathscr{M}(R)$ is homeomorphic to the Stone-Čech compactification of the discrete space $I$.

*Proof.* Yang: To be added. □

**Remark 1.13.** Here we list some facts about ultrafilters and the Stone-Čech compactification that will be used in the proof of Lemma 1.12.

Let $I$ be a set. A *filter* on $I$ is a nonempty collection $\mathscr{F} \subset \mathcal{P}(I)$ of subsets of $I$ such that:

- if $A, B \in \mathscr{F}$, then $A \cap B \in \mathscr{F}$;
- if $A \in \mathscr{F}$ and $A \subset B \subset I$, then $B \in \mathscr{F}$.

For example, let $I = \mathbb{N}$ and

$$\mathscr{F} = \{A \subset \mathbb{N} : \#(\mathbb{N} \setminus A) < \infty\}.$$

Then $\mathscr{F}$ is a filter on $\mathbb{N}$, called the *Fréchet filter*. Another example is the filter called the *principal filter* generated by a point $i_0 \in I$:

$$\mathscr{F}_{i_0} = \{A \subset I : i_0 \in A\}.$$

Filters are partially ordered by inclusion. A filter $\mathscr{U}$ is called an *ultrafilter* if it is maximal with respect to inclusion. A filter is an ultrafilter if and only if for every subset $A \subset I$, either $A \in \mathscr{U}$ or $I \setminus A \in \mathscr{U}$.

One can define the notion of convergence along ultrafilters in a topological space. Let $X$ be a topological space, $(x_i)_{i\in I}$ a collection of points in $X$ indexed by a set $I$, and $\mathscr{U}$ an ultrafilter on $I$. We say that $(x_i)_{i\in I}$ *converges* to a point $x \in X$ along the ultrafilter $\mathscr{U}$ if for every open neighborhood $U$ of $x$, the set $\{i \in I : x_i \in U\}$ belongs to $\mathscr{U}$. It is denoted by $\lim_{\mathscr{U}} x_i = x$. When $I = \mathbb{N}$ and $\mathscr{U}$ is the Fréchet filter, this notion of convergence coincides with the usual notion of convergence of sequences.

The Stone-Čech compactification of a discrete space is the largest compact Hausdorff space in which the original space can be densely embedded. Yang: To be checked.

**Theorem 1.14.** Let $R$ be a Banach ring. The spectrum $\mathscr{M}(R)$ is a compact Hausdorff space.

*Proof.* Yang: To be added. □

**Corollary 1.15.** Let $R$ be a Banach ring and $R^u$ its uniformization. The canonical map $\mathscr{M}(R^u) \to \mathscr{M}(R)$ induced by the bounded homomorphism $R \to R^u$ is a homeomorphism.

**Proposition 1.16.** Let $\mathbf{l}/\mathbf{k}$ be a finite normal extension of complete valuation fields and $A$ a Banach $\mathbf{k}$-algebra. Then we have:

(a) if $\mathbf{l}/\mathbf{k}$ is Galois, then $\mathscr{M}(A \otimes_{\mathbf{k}} \mathbf{l})$ admits a natural action of the Galois group $\mathrm{Gal}(\mathbf{l}/\mathbf{k})$ and the quotient space is homeomorphic to $\mathscr{M}(A)$;

(b) if $\mathbf{l}/\mathbf{k}$ is purely inseparable, then the canonical map $\mathscr{M}(A \otimes_{\mathbf{k}} \mathbf{l}) \to \mathscr{M}(A)$ is a homeomorphism.

*Proof.* Yang: To be added. □

**Corollary 1.17.** Let $\mathbf{k}$ be a complete valuation field, $\mathbb{k}$ its completion of an algebraic closure, and $A$ a Banach $\mathbf{k}$-algebra. Then we have a natural homeomorphism $\mathscr{M}(A \hat{\otimes}_{\mathbf{k}} \mathbb{k}) / \mathrm{Gal}(\mathbb{k}/\mathbf{k}) \cong \mathscr{M}(A)$.

*Proof.* Yang: To be added. □

## 1.2 Reduction map and kernel map

**Proposition 1.18.** Let $R$ be a Banach ring. The kernel map $\mathscr{M}(R) \to \mathrm{Spec}(R), x \mapsto \wp_x$ is continuous with respect to the Zariski topology on $\mathrm{Spec}(R)$.

*Proof.* Let $D(f) = \{f \neq 0\} \subset \mathrm{Spec}(R)$ be a principal open subset for some $f \in R$. The preimage of $D(f)$ under the kernel map is just the set $\{x \in \mathscr{M}(R) : |f|_x > 0\} = f^{-1}(\mathbb{R}_{>0})$, which is open in $\mathscr{M}(R)$ by definition of the topology on $\mathscr{M}(R)$. □

**Example 1.19.** Let us consider the spectrum $\mathscr{M}(\mathbb{Z})$ in Example 1.3. Under the kernel map $\mathscr{M}(\mathbb{Z}) \to \mathrm{Spec}(\mathbb{Z})$, the points $|\cdot|_p^\infty$ for each prime number $p$ are mapped to the prime ideal $(p)$, the other above points are all mapped to the zero ideal $(0)$.

Yang: Is this surjective? what is its fiber?

Yang: Seems i really need the surjectiviy to analysis spectrum of $\Bbbk\{\underline{T}\}$.

**Proposition 1.20.** Let $A$ be an $\mathbf{k}$-affinoid algebra over a complete non-archimedean field $\mathbf{k}$. Then the kernel map $\mathscr{M}(A) \to \mathrm{Spec}(A)$ is surjective.

*Proof.* Yang: To be add. ☐

**Construction 1.21.** Suppose that $R$ is a non-archimedean Banach ring with valuation subring $R^\circ$ and maximal ideal $R^{\circ\circ}$. For each $x \in \mathscr{M}(R)$, there is an induced homomorphism $R^\circ \to \mathscr{H}(x)^\circ$ between the valuation subrings. Furthermore, we have an induced homomorphism between the residue rings $\widetilde{R} = R^\circ/R^{\circ\circ} \to \mathscr{k}_{\mathscr{H}(x)}$. This gives rise to the *reduction map*

$$\mathrm{Red} : \mathscr{M}(R) \to \mathrm{Spec}(\widetilde{R}), \quad x \mapsto \ker(\widetilde{R} \to \mathscr{k}_{\mathscr{H}(x)}).$$

Let $f : R \to S$ be a bounded homomorphism of non-archimedean Banach rings. Yang: The following diagram commutes:

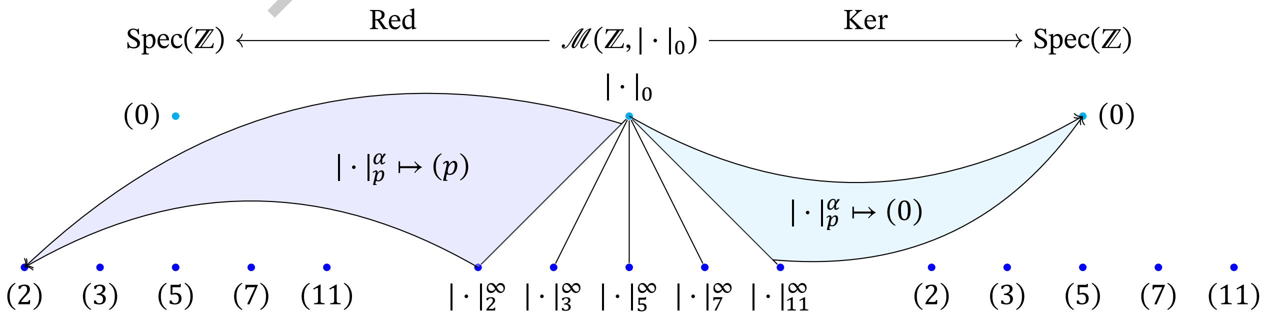
$$\begin{tikzcd}
\mathscr{M}(S) \arrow[r, "\mathrm{Red}"] \arrow[d, "\mathscr{M}(f)"'] & \mathrm{Spec}(\tilde{S}) \arrow[d, "\mathrm{Spec}(\tilde{f})"] \\
\mathscr{M}(R) \arrow[r, "\mathrm{Red}"] & \mathrm{Spec}(\widetilde{R})
\end{tikzcd}$$

Hence the reduction map is functorial with respect to bounded homomorphisms of non-archimedean Banach rings.

**Example 1.22.** Let $(\mathbb{Z}, |\cdot|_0)$ be the Banach ring with the trivial norm. The reduction ring is $\widetilde{\mathbb{Z}} = \mathbb{Z}$.

- $x = |\cdot|_p^\alpha$ for some prime number $p$ and $\alpha \in (0, \infty]$: $\mathscr{k}_{\mathscr{H}(x)} \cong \mathbb{F}_p$ and the induced homomorphism $\widetilde{\mathbb{Z}} = \mathbb{Z} \to \mathscr{k}_{\mathscr{H}(x)} = \mathbb{F}_p$ is the natural projection $\mathbb{Z} \to \mathbb{F}_p$;
- $x = |\cdot|_0$: $\mathscr{k}_{\mathscr{H}(x)} \cong \mathbb{Q}$ and the induced homomorphism $\widetilde{\mathbb{Z}} = \mathbb{Z} \to \mathscr{k}_{\mathscr{H}(x)} = \mathbb{Q}$ is the natural inclusion $\mathbb{Z} \to \mathbb{Q}$.

The following diagram illustrates the reduction map and the kernel map for the spectrum $\mathscr{M}(\mathbb{Z}, |\cdot|_0)$:




**Proposition 1.23.** Let $R$ be a non-archimedean Banach ring and $\widetilde{U} \subset \mathrm{Spec}(\widetilde{R})$ be a Zariski open subset. Then the preimage $\mathrm{Red}^{-1}(\widetilde{U})$ is a closed subset of $\mathscr{M}(R)$.

*Proof.* Yang: To be completed. □

**Proposition 1.24.** Let $\mathbf{k}$ be a complete non-archimedean field and $A, B$ a strictly $\mathbf{k}$-affinoid algebras. Suppose that $\varphi : A \to B$ is a bounded $\mathbf{k}$-algebra homomorphism. Then we have:

(a) if $\varphi$ is finite, then the induced homomorphism $\widetilde{\varphi} : \widetilde{A} \to \widetilde{B}$ is finite;

(b) if $\varphi$ is of finite type, then the induced homomorphism $\widetilde{\varphi} : \widetilde{A} \to \widetilde{B}$ is of finite type.

Yang: To be revised.

## 1.3 Spectrum of Tate algebras

In this subsection, fix an algebraically closed complete non-archimedean field $\Bbbk$.

**Spectrum of Tate algebra in one variable** Let $A = \Bbbk\{T/r\}$. We list some types of points in the spectrum $\mathscr{M}(A)$.

**Construction 1.25.** For each $a \in \Bbbk$ with $|a| \le r$, we have the *type I* point $x_a$ corresponding to the evaluation at $a$, i.e., $|f|_{x_a} \coloneqq |f(a)|$ for each $f \in A$.

For each closed disk $E = E(a, s) \coloneqq \{b \in \Bbbk : |b - a| \le s\}$ with center $a \in \Bbbk$ and radius $s \le r$, we have the point $x_E = x_{a,s}$ corresponding to the multiplicative semi-norm defined by

$$|f|_{x_E} = |f|_{x_{a,s}} \coloneqq \sup_{b \in E(a,s)} |f(b)|$$

for each $f \in A$. If $s \in |\Bbbk^\times|$, then the point $x_E$ is called a *type II* point; otherwise, it is called a *type III* point.

Let $E_n = E(a_n, s_n)$ be a sequence of closed disks in $\Bbbk$ such that $E_{n+1} \subsetneq E_n$ and $\bigcap_n E_n = \varnothing$. Then we have the point $x_{\{E_n\}} = x_{\{a_n, s_n\}}$ corresponding to the multiplicative semi-norm defined by

$$|f|_{x_{\{E_n\}}} = |f|_{x_{\{a_n,s_n\}}} \coloneqq \inf_n |f|_{x_{E_n}}$$

for each $f \in A$. Such a point is called a *type IV* point.

**Proposition 1.26.** The points in the spectrum $\mathscr{M}(\Bbbk\{r^{-1}T\})$ can be classified into four types as described in Construction 1.25.

*Proof.* Fix $x \in \mathscr{M}(\Bbbk\{r^{-1}T\})$, set

$$s = \inf_{a \in \Bbbk} |T - a|_x \le r, \quad E = \{a \in \Bbbk : |T - a|_x = s\} \subset E(0, r).$$

**Case 1.** $E \neq \varnothing$ and $s = 0$.

By assumption, there exists $a \in E$ such that $|T - a|_x = 0$. Note that if $f(a) = 0$, then $T - a \mid f$ in $\Bbbk\{r^{-1}T\}$ and hence $|f|_x = |T - a|_x |g|_x = 0$. Then we have

$$|f(a)| = ||f(a)| - |f(a) - f|_x| \le |f|_x \le |f(a)| + |f(a) - f|_x = |f(a)|$$

for each $f \in \Bbbk\{r^{-1}T\}$, which implies that $|f|_x = |f(a)|$. Thus $x$ is a type I point $x_a$.

**Case 2.** $E \neq \varnothing$ and $s > 0$.

Let $a \in E$. Note that for every $b \in \Bbbk$, we have

$$|a-b| \leq \max\{|T-a|_x, |T-b|_x\} = |T-b|_x.$$

First we show that $E = E(a,s)$. For each $b \in E(a,s)$, we have $|T-b|_x \leq \max\{|T-a|_x, |a-b|\} = s$, which implies that $b \in E$. Conversely, for each $b \in E$, we have $|a-b| \leq \max\{|T-a|_x, |T-b|_x\} = s$.

Let $f \in \Bbbk[T]$ be a polynomial. Write $f = \prod_{i=1}^n (T-c_i)$ for some $c_1, \dots, c_n \in \Bbbk$. Then we have

$$|f|_x = \prod_{i=1}^n |T-c_i|_x \geq \prod_{i=1}^n |b-c_i| = |f(b)|, \quad \forall b \in E.$$

I claim that for every $\varepsilon \in (0,1)$, there exists $b \in E$ such that $\varepsilon|T-c_i|_x < |b-c_i|$ for each $i = 1, \dots, n$. Indeed, if $c_i \notin E$, then $|T-c_i|_x = |b-c_i|$ for each $b \in E$. Hence we only need to consider the case when $c_i \in E$. Since $\Bbbk$ is algebraically closed, $E(a,s) \setminus \bigcup_{i=1}^n E(c_i, \varepsilon s) \neq \varnothing$. Choose $b$ in the set. Then we have

$$|f(b)| \geq \prod_{i=1}^n \varepsilon|T-c_i|_x = \varepsilon^n |f|_x.$$

Thus $|f|_x = \sup_{b \in E} |f(b)|$ for each polynomial $f \in \Bbbk[T]$. Since polynomials are dense in $\Bbbk\{r^{-1}T\}$, we have $|f|_x = \sup_{b \in E} |f(b)|$ for each $f \in \Bbbk\{r^{-1}T\}$. Therefore, $x$ is the point $x_E = x_{a,s}$, which is of type II or type III depending on whether $s \in |\Bbbk^\times|$ or not.

**Case 3.** $E = \varnothing$.

Set $E_n = \{a \in \Bbbk : |T-a|_x \leq s + 1/n\}$ and $a_n \in E_n$ for each $n \in \mathbb{N}$. By the similar argument as in Case 2, we have $E_n = E(a_n, s+1/n)$. Note that $E_n$ is a decreasing sequence of closed disks with $\bigcap_n E_n = E = \varnothing$.

For $c \in \Bbbk$, there exists $N$ such that $\forall n \geq N$, we have

$$c \notin E_n \implies |T-c|_x > |T-a_n|_x \implies |T-c|_x = |a_n - c|.$$

Thus

$$\inf_n |T-c|_{E_n} = \inf_n |a_n - c| = |T-c|_x.$$

By multiplicativity, we have $\inf_n |f|_{E_n} = |f|_x$ for each polynomial $f \in \Bbbk[T]$. And then by density of polynomials, the equality holds for each $f \in \Bbbk\{r^{-1}T\}$. Therefore, $x = x_{\{E_n\}} = x_{\{a_n, s_n\}}$ is of type IV. □

Now we describe the structure of $X = \mathscr{M}(\Bbbk\{T\})$ in more details. For each $a \in \Bbbk$ with $|a| \leq 1$, there is a line segment $L_{\xi, x_a} = \{x_{a,r} | r \in [0,1]\}$ in $X$ connecting the Gauss point $x_{0,1}$ and the type I point $x_a$. Two such line segments $L_{\xi, x_a}$ and $L_{\xi, x_b}$ will meet at the point $x_{a,s} = x_{b,s}$ where $s = |a-b|$.

On such a line segment $L_{\xi, x_a}$, if $s \in |\Bbbk^\times|$, then the point $x_{a,s}$ is of type II; otherwise, it is of type III. One can think of the type II points like rational points and the type III points like irrational points on the line segment.

On a type II point $x_{a,s}$, there are infinitely many line segments $L_{x_{a,s}, x_b}$ connecting $x_{a,s}$ and type I points $x_b$ with $|b-a| < s$. It looks like a tree with infinitely many branches rooted at $x_{a,s}$. The branch is indexed by the quotient set $E(a,s)/B(a,s) \cong \mathscr{k}_{\Bbbk}$. On a type III point $x_{a,s}$, there are no line segments branching out. In a slogan, the spectrum $\mathscr{M}(\Bbbk\{T\})$ looks like a fractal tree with infinitely many branches. In the tree, $|\Bbbk^\times|$ controls the placement of roots of branches, while $\mathscr{k}_{\Bbbk}$ controls the number of branches at each root.

Consider a sequence of disks $E_n = E(a_n, s_n)$ such that $E_{n+1} \subset E_n$. It corresponds to a piecewise linear path $L_{x_{a_1,s_1}, x_{a_2,s_2}} \cup L_{x_{a_2,s_2}, x_{a_3,s_3}} \cup \cdots$ in the tree. Topologically, it is homeomorphic to $[0,1)$. If the intersection $\bigcap_n E_n$ is nonempty, then it corresponds to a type I,II, or III point fitting into the place of $1$ in the path. It is compact and everything is fine. However, if the intersection $\bigcap_n E_n$ is empty, then it will look like $[0,1)$ with a missing point $1$ locally if we only consider the points of type I, II, and III. The type IV point $x_{\{E_n\}}$ is exactly the missing point $1$ in the path, which makes the path compact.

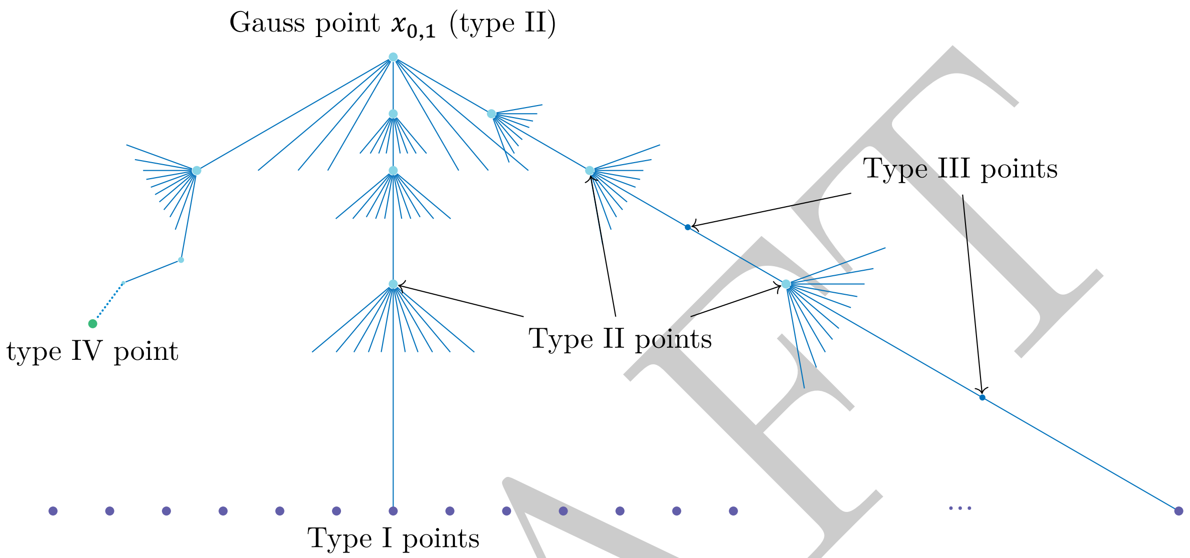



Figure 1: The Berkovich spectrum $\mathscr{M}(\Bbbk\{T\})$

**Proposition 1.27.** The completed residue fields of the four types of points in the spectrum $\mathscr{M}(\mathbf{k}\{r^{-1}T\})$ are described as follows:

- type I point $x_a$: $\mathscr{H}(x_a)$ is isomorphic to $\Bbbk$;
- type II point $x_{a,s}$: $\mathscr{H}(x_{a,s}) \cong \mathscr{k}_{\Bbbk}((t))$;
- type III point $x_{a,s}$: $\mathscr{k}_{\mathscr{H}(x_{a,s})} \cong \mathscr{k}_{\Bbbk}$ and the value group $|\mathscr{H}(x_{a,s})^{\times}|$ is generated by $|\Bbbk^{\times}|$ and $s$;
- type IV point $x_{\{a_n,s_n\}}$: $\mathscr{H}(x_{\{a_n,s_n\}})$ is an immediate extension of $\Bbbk$.

Yang: To be checked.

*Proof.* Yang: To be completed. □

**Example 1.28.** The completed residue field $\mathscr{H}(x_a)$ for a type I point $x_a$ with $a \in \Bbbk$ and $|a| \le r$ is isomorphic to $\Bbbk$. Yang: To be complete.

**Example 1.29.** Here we describe the reduction map and the kernel map for the spectrum $\mathscr{M}(A)$. The reduction ring is

$$\widetilde{A} = \begin{cases} \mathscr{k}_{\Bbbk}[T], & \text{if } r \in |\Bbbk^{\times}|; \\ \mathscr{k}_{\Bbbk}, & \text{if } r \notin |\Bbbk^{\times}|. \end{cases}$$

If $r \notin |\Bbbk^{\times}|$, then the reduction map $\mathsf{Red} : \mathscr{M}(A) \to \mathsf{Spec}(\widetilde{A}) = \mathsf{Spec}(\mathscr{k}_{\Bbbk})$ is trivial, i.e., every point in

$\mathscr{M}(A)$ is mapped to the zero ideal of $\mathscr{k}_{\Bbbk}$. Hence we only consider the case when $r \in |\Bbbk^\times|$. Without loss of generality, we may assume that $r = 1$.

For each type I point $x_a$ with $a \in \Bbbk$ and $|a| \leq 1$, the induced homomorphism $\widetilde{A} = \mathscr{k}_{\Bbbk}[T] \to \mathscr{k}_{\mathscr{H}(x_a)} = \mathscr{k}_{\Bbbk}$ is given by $T \mapsto a \mod \Bbbk^{\circ\circ}$. Thus the reduction map $\mathsf{Red} : \mathscr{M}(A) \to \mathsf{Spec}(\widetilde{A}) = \mathsf{Spec}(\mathscr{k}_{\Bbbk}[T])$ sends the type I point $x_a$ to the maximal ideal $(T - \tilde{a})$. For a type II or type III point $x_{a,s}$ with $a \in \Bbbk$ and $s \leq 1$, the homomorphism $A^\circ \to \mathscr{H}(x_{a,s})^\circ$

Yang: To be continued.

**Example 1.30.** Here we describe the kernel map for the spectrum $\mathscr{M}(A)$. If $x = x_a$ is a type I point corresponding to $a \in \Bbbk$ with $|a| \leq r$, then $\wp_{x_a} = (T - a)$. For type II, type III or type IV points $x$, note that if $f \in A$ is invertible, then $|f|_x > 0$ for each $x \in \mathscr{M}(A)$ since $|f|_x |f^{-1}|_x = |1| = 1$. By Yang: ref, we only need to consider elements in $A$ of the form $T - c$ for some $c \in \Bbbk$ with $|c| \leq r$. Since $|T - c|_x \neq 0$ for all $c \in \Bbbk$ with $|c| \leq r$, we have $\wp_x = (0)$. To summarize, the kernel map $\mathscr{M}(A) \to \mathsf{Spec}(A)$ is given by

$$\wp_x = \begin{cases} (T - a), & \text{if } x = x_a \text{ is a type I point;} \\ (0), & \text{if } x \text{ is a type II, type III or type IV point.} \end{cases}$$

**Spectrum of Tate algebra in servel variables** Let $A = \Bbbk\{\underline{T}\}$. To consider the spectrum $\mathscr{M}(A)$, inspired by Example 1.30, we first look at the image of the points in $\mathscr{M}(A)$ under the kernel map $\mathscr{M}(A) \to \mathsf{Spec}(A)$.

For every $\mathfrak{p} \in \mathsf{Spec}(A)$, since $A$ is noetherian, $\mathfrak{p}$ is finitely generated, say $\mathfrak{p} = (f_1, \dots, f_n)$. On $E(0,1) \subset \Bbbk^n$, $\mathfrak{p}$ defines a closed analytic subspace $V = V(f_1, \dots, f_n) = \{x \in E(0,1) : f_i(x) = 0, i = 1, \dots, n\}$. By Yang: ref, $V$ is nonempty. On $E(0,1)$, we have the supremum norm $\|\cdot\|_{\mathsf{sup}}$. It inherits the metric from $E(0,1)$ and hence is a complete metric space.

According to the proof of Proposition 1.26, we should use prime divisor on $\mathsf{Spec}(A/\mathfrak{p})$ to replace the role of points in one variable case.

Yang: To be continued.

# 2 Affinoid spaces

Fix a non-archimedean completed valued field $(\mathbf{k}, |\cdot|)$ throughout this section. Let $A$ be a $\mathbf{k}$-affinoid algebra, and let $X = \mathscr{M}(A)$ be the associated affinoid space. It should be viewed as the non-archimedean analogue of an affine variety in algebraic geometry or an affine scheme in scheme theory. The goal of this section is to define a suitable Grothendieck topology on $X$ such that we can define a structure sheaf of analytic functions on $X$, and then study its basic properties.

## 2.1 Affinoid domains

On a general open subset $U \subseteq X$, the ring of analytic functions on $U$ may behave badly.

Yang: To be continued.

**Definition 2.1.** A closed subset $V \subseteq X$ is called an *affinoid domain* if there exists a $\mathbf{k}$-affinoid algebra $A_V$ and a morphism of $\mathbf{k}$-affinoid algebras $\theta : A \to A_V$ satisfying the following universal property: for every bounded homomorphism of $\mathbf{k}$-affinoid algebras $\psi : A \to B$ such that the induced map on spectra $\mathscr{M}(\psi) : \mathscr{M}(B) \to X$ has its image contained in $V$, there exists a unique bounded homomorphism $\varphi : A_V \to B$ such that the following right diagram commutes:

$$\begin{tikzcd} & V \arrow[dl, hook'] & \\ X & & \mathscr{M}(B) \arrow[ul] \arrow[ll, "\mathscr{M}(\psi)"] \end{tikzcd} \quad \rightsquigarrow \quad \begin{tikzcd} & A_V \arrow[dr, "\varphi"] & \\ A \arrow[ur, "\theta"] \arrow[rr, "\psi"'] & & B \end{tikzcd}$$

In this case, we say that $V$ is represented by the $\mathbf{k}$-affinoid algebra $A_V$. If $A_V$ is strict, then we say that $V$ is a strict affinoid domain.

**Slogan** *A closed subset $V \subset X$ is an affinoid domain if the functor "*$\mathrm{Mor}(-, V)$*" is representable by a $\mathbf{k}$-affinoid algebra $A_V$.*

There are several important classes of affinoid domains.

**Construction 2.2.** Let $f = (f_1, \dots, f_n), g = (g_1, \dots, g_m)$ be two tuples of elements in $A$ and $r = (r_1, \dots, r_n), s = (s_1, \dots, s_m)$ be two tuples of positive real numbers. Consider the following closed subset of $X$:

$$X\left(\underline{f/r}; \underline{s/g}\right) \coloneqq \{x \in X : \ |f_i(x)| \le r_i, |g_j(x)| \ge s_j, 1 \le i \le n, 1 \le j \le m\}.$$

Such a closed subset is called a *Laurent domain* of $X$. If $m = 0$, then it is called a *Weierstrass domain* of $X$, denoted by

$$X\left(\underline{f/r}\right) \coloneqq \{x \in X : |f_i(x)| \le r_i, 1 \le i \le n\}.$$

More generally, let $f = (f_1, \dots, f_n), g$ be elements in $A$ such that the ideal generated by them is the whole algebra $A$. Set $p = (p_1, \dots, p_n)$ be a tuple of positive real numbers. We define the following closed subset of $X$:

$$X\left(\underline{(f/g)/p}\right) \coloneqq \{x \in X : \ |f_i(x)| \le p_i |g(x)|, 1 \le i \le n\}.$$

Such a closed subset is called a *rational domain* of $X$.

**Proposition 2.3.** Let $A$ be a $\mathbf{k}$-affinoid algebra, and let $X = \mathscr{M}(A)$ be the associated affinoid space. Then Weierstrass domains, Laurent domains, and rational domains of $X$ are affinoid domains. They are represented by $A\{f/r\}$, $A\{f/r; g/s\}$, and $A\{(f/g)/p\}$ in Yang: ref, respectively.

*Proof.* Yang: To be added. □

We have a sequence of inclusion:

$$\{\text{Weierstrass domains}\} \subseteq \{\text{Laurent domains}\} \subseteq \{\text{Rational domains}\} \subseteq \{\text{Affinoid domains}\}.$$

Yang: Why Laurent domain is rational domain?

Yang: Is these domain open? Yang: In the sence of Tate, they are.

**Proposition 2.4.** Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$-affinoid algebra $A_V$. Then the natural map on spectra $\mathscr{M}(\theta) : \mathscr{M}(A_V) \to X$ induced by the representing homomorphism $\theta : A \to A_V$ has its image equal to $V$, and induces a homeomorphism $\mathscr{M}(A_V) \cong V$.

*Proof.* Yang: To be added. □

**Proposition 2.5.** Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$-affinoid algebra $A_V$. Then the natural bounded homomorphism $\varphi : A \to A_V$ is flat.

*Proof.* Yang: To be added. □

**Proposition 2.6.** Let $f : Y = \mathscr{M}(B) \to X = \mathscr{M}(A)$ be a morphism of $\mathbf{k}$-affinoid spaces induced by a bounded homomorphism of $\mathbf{k}$-affinoid algebras $\phi : A \to B$. Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$-affinoid algebra $A_V$. Then the inverse image $f^{-1}(V) \subseteq Y$ is an affinoid domain represented by the $\mathbf{k}$-affinoid algebra $B\hat{\otimes}_A A_V$. Yang: To be revised.

*Proof.* Yang: To be added. □

**Proposition 2.7.** Let $V_1, V_2 \subseteq X$ be two affinoid domains represented by the $\mathbf{k}$-affinoid algebras $A_{V_1}, A_{V_2}$, respectively. Then the intersection $V_1 \cap V_2 \subseteq X$ is an affinoid domain represented by the $\mathbf{k}$-affinoid algebra $A_{V_1}\hat{\otimes}_A A_{V_2}$. Yang: To be checked.

*Proof.* Yang: To be added. □

**Proposition 2.8.** Let $x \in X$ be a point. Then the collection of strict affinoid domains of $X$ containing $x$ forms a fundamental system of neighborhoods of $x$. Explicitly, for any open neighborhood $U \subseteq X$ of $x$, there exists a strict affinoid domain $V \subseteq X$ such that $x \in V \subseteq U$. Yang: To be checked.

**Theorem 2.9** (Gerritzen-Grauert theorem)**.** Let $V \subseteq X$ be an affinoid domain. Then there exist finitely many rational domains $V_1, \ldots, V_m \subseteq X$ such that $V = V_1 \cup \cdots \cup V_m$. Yang: To be checked.

*Proof.* Yang: To be added. □

**Definition 2.10.** Let $V \subseteq X$ be an affinoid domain represented by the $\mathbf{k}$-affinoid algebra $A_V$. Suppose that $A_V\hat{\otimes}\mathbf{K}$ is strictly $\mathbf{K}$-affinoid for some completed valued field extension $\mathbf{K}/\mathbf{k}$. The *dimension* of $V$ is defined as the Krull dimension of the affinoid algebra $A_V\hat{\otimes}\mathbf{K}$:

$$\dim(V) := \dim_{\mathrm{Krull}}(A_V\hat{\otimes}\mathbf{K}).$$

By ??, the definition is independent of the choice of $\mathbf{K}$.

## 2.2 The Grothendieck topology and structure sheaf

**Theorem 2.11** (Tate's Acyclicity Theorem)**.** Let $A$ be a $\mathbf{k}$-affinoid algebra, and let $X = \mathscr{M}(A)$ be the associated affinoid space. Let $\{V_i\}_{i\in I}$ be a finite covering of $X$ by affinoid domains. For each finite subset $J \subseteq I$, set $V_J := \bigcap_{j\in J} V_j$, which is an affinoid domain by Proposition 2.7. Then the augmented Čech complex

$$0 \to A \to \prod_{i\in I} A_{V_i} \to \prod_{\substack{J\subseteq I\\|J|=2}} A_{V_J} \to \prod_{\substack{J\subseteq I\\|J|=3}} A_{V_J} \to \cdots$$

is exact. Yang: To be checked.

*Proof.* Yang: To be added. □

**Definition 2.12.** A *special domain* of $X$ is a finite union of affinoid domains of $X$. Let $V \subseteq X$ be a special domain, say $V = V_1 \cup \cdots \cup V_m$ with each $V_i$ an affinoid domain. We define

$$A_V := \ker\left(\prod_{i=1}^{m} A_{V_i} \to \prod_{1\le i<j\le m} A_{V_i\cap V_j}\right).$$

Yang: To be checked.

**Proposition 2.13.** Let $V \subseteq X$ be a special domain, say $V = V_1 \cup \cdots \cup V_m$ with each $V_i$ an affinoid domain. Then $V$ is an affinoid domain if and only if the $\mathbf{k}$-algebra $A_V$ defined in Definition 2.12 is a $\mathbf{k}$-affinoid algebra. In this case, $V$ is represented by the $\mathbf{k}$-affinoid algebra $A_V$. Yang: To be added.

**Definition 2.14.** The *site of special domains* on $X$, is defined as follows:

- the underlying category has as objects the special domains $V \subseteq X$, and the morphisms are given by the inclusion maps;
- a covering of an objects $V \subseteq X$ is a finite family of affinoid domains $\{V_i\}_{i\in I}$ such that $V = \bigcup_{i\in I} V_i$.

The *structure sheaf on site of special domains* is given by assigning to each special domain $V \subseteq X$ the ring $A_V$. Yang: To be revised.

**Definition 2.15.** For every open subset $U \subseteq X$, we define the ring of analytic functions on $U$ as follows:

$$\mathscr{O}_X(U) := \varprojlim_{V\subseteq U} A_V \quad \in \mathbf{Alg}_{\mathbf{k}},$$

where the limit is taken over all special domains $V \subseteq U$, and $A_V$ is the $\mathbf{k}$-algebra representing the special domain $V$. Yang: To be revised.

**Remark 2.16.** Note that in Definition 2.14 we have defined a site on $X$ consisting of special domains, and in Definition 2.15 we have defined a structure sheaf on the usual topological space $X$. These two sheaves are closely related, and also has some differences.

Yang: on special site, the assigning algebra is a normed algebra, $\mathbf{k}$-affinoid algebra. On open subset, the assigning algebra is even not necessarily normed.

Yang: If a special subset $V \subseteq X$ is open, then they coincide?

Yang: On a point, they are the same?Yang: Yes, this is right.

Yang: To be revised.

**Definition 2.17.** The morphism of $\mathbf{k}$-affinoid spaces is defined as the morphism $(f, f^\sharp)$ of locally ringed spaces satisfying that

(a) the underlying map $f : Y \to X$ is continuous with respect to the spectral topology;

(b) the morphism $f^\sharp : \mathscr{O}_X \to f_* \mathscr{O}_Y$ is bounded, that is, for every special domains $V \subseteq X$ and $W \subseteq Y$ such that $f(W) \subseteq V$, the induced homomorphism $A_V \to A_W$ is bounded.

We denote by Yang: the category of $\mathbf{k}$-affinoid spaces with morphisms defined as above.

**Proposition 2.18.** The category of $\mathbf{k}$-affinoid spaces is equivalent to the category of $\mathbf{k}$-affinoid algebras with bounded homomorphisms via the functor $\mathscr{M}$.

Let $M$ be an $A$-module, we can associate to $M$ a sheaf $\mathscr{O}_X(M)$ of $\mathscr{O}_X$-modules on $X$ by setting

$$\Gamma(U, \mathscr{O}_X(M)) := \varprojlim_{V \subseteq U} A_V \otimes_A M,$$

where the limit is taken over all special domains $V \subseteq U$.

**Proposition 2.19.** The functor $M \mapsto \mathscr{O}_X(M)$ induces an equivalence of categories between the category of finite $A$-modules and the category of coherent $\mathscr{O}_X$-modules on $X$.

## 2.3 Boundary and interior

Let $A, B$ be two $\mathbf{k}$-affinoid algebras, and let $f : Y = \mathscr{M}(B) \to X = \mathscr{M}(A)$ be a morphism of $\mathbf{k}$-affinoid spaces induced by a bounded homomorphism of $\mathbf{k}$-affinoid algebras $\phi : A \to B$.

**Definition 2.20.** The *relative interior* of $Y$ over $X$ is defined as

$$(Y/X) := \{y \in Y : \text{there exists an admissible surjective} A \to B \text{ inducing } y\}.$$

Yang: To be revised.

**Definition 2.21.**

## 2.4 Some local properties

In this subsection, we will study some local properties of affinoid spaces. As in algebraic geometry, the local ring at a point $x \in X$ is defined as

$$\mathscr{O}_{X,x} := \varinjlim_{x \in U} \mathscr{O}_X(U),$$

where the limit is taken over all open neighborhoods $U$ of $x$ in $X$.

**Proposition 2.22.** We have

$$\mathscr{O}_{X,x} \cong \varinjlim_{\substack{x \in V \\ \text{affinoid}}} A_V \cong \varinjlim_{\substack{x \in V \\ \text{special}}} A_V.$$

Yang:

**Theorem 2.23.** The ring $\mathscr{O}_{X,x}$ at any point $x \in X$ is a noetherian local ring. And it is faithfully flat over the local ring $\mathcal{O}_{\operatorname{Spec} A, \wp_x}$.

*Proof.* Yang: To be added. □

In algebraic geometry, there is an amazing fact we often overlook: the residue field of the local ring at a point $x$ of an affine variety is isomorphic to the residue field of the local ring at $x$ of any open neighborhood of $x$. This fails in the non-archimedean analytic geometry, see Example 2.24.

**Example 2.24.** Let $A = \Bbbk\{T\}$ and $X = \mathscr{M}(A)$.

Yang: To be added.

By Example 2.24, the residue field $\operatorname{Frac} A/\wp_x$ may not be a good invariant of the point $x$. This seems a good reason to consider the completed residue field $\mathscr{H}(x)$ instead, by Proposition 2.25.

**Proposition 2.25.** Let $x \in X$ be a point, and let $V$ be an affinoid domain containing $x$ represented by the $\mathbf{k}$-affinoid algebra $A_V$. Then the completed residue field $\mathscr{H}(x)$ is isomorphic to the completion of the residue field $\operatorname{Frac} A_V/\wp_x$ with respect to the norm induced by the point $x$.
Furthermore, the completed residue field $\mathscr{H}(x)$ contains the residue field $\mathscr{O}_{X,x}/\mathfrak{m}_x$ as a dense subfield.

*Proof.* Yang: To be added. □

**Definition 2.26.** Let $\mathcal{P}$ be one of the following properties: regular, normal, reduced, Gorenstein, Cohen-Macaulay, complete intersection, etc. We say that the affinoid space $X$ has a local property $\mathcal{P}$ at a point $x \in X$ if the local ring $\mathscr{O}_{X,x}$ has the property $\mathcal{P}$. Yang: To be added.

Using $\mathcal{P}(X)$ to denote the set of points $x \in X$ such that $X$ has the property $\mathcal{P}$ at $x$, we have the following result.

**Theorem 2.27.** Let $X$ be an affinoid space, and let $\mathcal{P}$ be one of the following properties: regular, normal, reduced, Gorenstein, Cohen-Macaulay, complete intersection. Then the locus of points $x \in X$ such that $X$ has the property $\mathcal{P}$ at $x$ is a Zariski open subset of $X$. More explicitly, we have $\mathcal{P}(X) = \operatorname{Ker}^{-1}(\mathcal{P}(\operatorname{Spec} A))$.

# 3 Analytic spaces

Fix a complete non-archimedean field $\mathbf{k}$.

## 3.1 Definition of analytic spaces

**Definition 3.1.** Let $X$ be a topological space. A *net* on $X$ is a collection $\mathcal{N}$ of subsets of $X$ such that

(a) for every $x \in X$, there exist $U_1, \dots, U_n \in \mathcal{N}$ such that $x \in \bigcap_i U_i$ and $\bigcup_i U_i$ is an open neighborhood of $x$ in $X$;

(b) for every $U, V \in \mathcal{N}$, the collection $\mathcal{N}|_{U\cap V} := \{W \in \mathcal{N} : W \subseteq U \cap V\} \subset \mathcal{P}(U \cap V)$ satisfies the first condition for the topological space $U \cap V$.

Yang: For example, if $X$ is a **k**-affinoid space, then the collection of all affinoid domains in $X$ forms a net on $X$.

**Definition 3.2.** Let $X$ be a locally Hausdorff topological space. A **k**-*affinoid atlas* on $X$ is a net $\mathcal{N}$ on $X$ together with

(a) for every $U \in \mathcal{N}$, a structure of a **k**-affinoid space on $U$, i.e. a **k**-affinoid algebra $A_U$ and an isomorphism $U \cong \mathcal{M}(A_U)$;

(b) for every $U, V \in \mathcal{N}$ with $V \subseteq U$, a morphism $\mathrm{res}^U_V : A_U \to A_V$ of **k**-affinoid algebras such that the induced morphism $\mathcal{M}(A_V) \to \mathcal{M}(A_U)$ making $V$ a **k**-affinoid subdomain in $U$.

A **k**-*analytic space* or *Berkovich space* is a locally Hausdorff topological space $X$ together with a **k**-affinoid atlas on $X$.

**Remark 3.3.** There are two different versions of the definition of **k**-analytic spaces in the literature: one given by Berkovich in Yang:

**Definition 3.4.** Let $(X, \mathscr{O}_X)$ and $(Y, \mathscr{O}_Y)$ be **k**-analytic spaces. A *morphism* $f : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ of **k**-analytic spaces is a morphism of locally ringed spaces. The category of **k**-analytic spaces is the category whose objects are **k**-analytic spaces and whose morphisms are morphisms of **k**-analytic spaces. Yang: To be revised.

**Theorem 3.5.** Let $(X, \mathscr{O}_X)$, $(Y, \mathscr{O}_Y)$, and $(Z, \mathscr{O}_Z)$ be **k**-analytic spaces, and let $f : (X, \mathscr{O}_X) \to (Z, \mathscr{O}_Z)$ and $g : (Y, \mathscr{O}_Y) \to (Z, \mathscr{O}_Z)$ be morphisms of **k**-analytic spaces. Then the fibre product $(X, \mathscr{O}_X) \times_{(Z,\mathscr{O}_Z)} (Y, \mathscr{O}_Y)$ exists in the category of **k**-analytic spaces. Yang: To be revised.

**Definition 3.6.** Let $(X, \mathscr{O}_X)$ be a **k**-analytic space. We say that $(X, \mathscr{O}_X)$ is *good* if every point $x \in X$ has an open neighborhood $U$ such that $(U, \mathscr{O}_X|_U)$ is a **k**-affinoid space.

## 3.2 The first properties

**Definition 3.7.** Let $f : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of **k**-analytic spaces. We say that $f$ is *finite* if for every affinoid domain $V \subseteq Y$, the preimage $f^{-1}(V)$ is an affinoid domain in $X$ and the induced morphism $\mathscr{O}_Y(V) \to \mathscr{O}_X(f^{-1}(V))$ is a finite morphism of **k**-affinoid algebras. Yang:

**Theorem 3.8.** Let $f : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of **k**-analytic spaces. Then there exists a factorization $f = g \circ h$ where $h : (X, \mathscr{O}_X) \to (Z, \mathscr{O}_Z)$ is a finite morphism and $g : (Z, \mathscr{O}_Z) \to (Y, \mathscr{O}_Y)$ is a morphism such that for every affinoid domain $W \subseteq Y$, the preimage $g^{-1}(W)$ is an affinoid domain in $Z$ and the induced morphism $\mathscr{O}_Y(W) \to \mathscr{O}_Z(g^{-1}(W))$ is a flat morphism of **k**-affinoid algebras. Yang:

**Definition 3.9.** Let $f : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of **k**-analytic spaces. We say that $f$ is *separated* if the diagonal morphism $\Delta_f : (X, \mathscr{O}_X) \to (X, \mathscr{O}_X) \times_{(Y, \mathscr{O}_Y)} (X, \mathscr{O}_X)$ is a closed immersion. Yang:

**Definition 3.10.** Let $f : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of **k**-analytic spaces. We say that $f$ is *proper* if it is separated, compact, and without boundary. Yang:

**Theorem 3.11.** Let $f : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a proper morphism of **k**-analytic spaces. Then for every coherent $\mathscr{O}_X$-module $\mathscr{F}$, the direct image $f_*(\mathscr{F})$ is a coherent $\mathscr{O}_Y$-module. Yang:

**Definition 3.12.** Let $f : (X, \mathscr{O}_X) \to (Y, \mathscr{O}_Y)$ be a morphism of **k**-analytic spaces. We say that $f$ is *smooth* (resp. *étale*) if for every $x \in X$, there exist open neighborhoods $U$ of $x$ in $X$ and $V$ of $f(x)$ in $Y$ such that $f(U) \subseteq V$ and a commutative diagram Yang:

# 4 GAGA

In this section, we discuss the GAGA (Géométrie Algébrique et Géométrie Analytique) principle for Berkovich spaces.

**Construction 4.1.** Let $X$ be a scheme of finite type over a non-archimedean field $k$. We can associate to $X$ its Berkovich analytification $X^{\mathrm{an}}$, which is a Berkovich analytic space over $k$.

The analytification functor $(\cdot)^{\mathrm{an}}$ has the following properties:

- It is compatible with fiber products: for schemes $X$ and $Y$ over $k$, we have $(X \times_k Y)^{\mathrm{an}} \cong X^{\mathrm{an}} \times_k Y^{\mathrm{an}}$.
- It preserves open and closed immersions.
- It induces a homeomorphism between the underlying topological spaces of $X$ and $X^{\mathrm{an}}$.
- For a coherent sheaf $\mathcal{F}$ on $X$, there is an associated coherent sheaf $\mathcal{F}^{\mathrm{an}}$ on $X^{\mathrm{an}}$.

Yang: To be revised.